Adaptive Water Wisdom Engine (AWWE)

> This document outlines the end-to-end data architecture for the AWWE project, mapping how multi-modal data is ingested, processed, modeled, and served to stakeholders.

1. Data Sources & Ingestion Layer

This layer handles the collection of structured, semi-structured, and unstructured data from various external and internal sources.

- **Meteorological Data:** Real-time APIs and historical databases for rainfall, temperature, and weather forecasts (e.g., IMD data).
- **Geospatial & Satellite Data:** Digital Elevation Models (DEM) and Land-Use/Land-Cover (LULC) imagery via satellite remote sensing (e.g., Sentinel, Landsat).
- **Hydrological Gauge Records:** Historical water level, inflow, and outflow records from the Water Resources Department (WRD).
- **Traditional Ecological Knowledge (TEK):** Qualitative community wisdom, local tank management rules, and historical failure points gathered via surveys or field notes.

2. Data Storage Layer

Given the multi-modal nature of the project, a hybrid storage approach is recommended.

- **Raw Data Lake:** (e.g., Amazon S3, Google Cloud Storage) Stores raw satellite images, unstructured TEK text, and raw JSON/CSV dumps of weather data.
- **Relational/Spatial Database:** (e.g., PostgreSQL with PostGIS) Stores structured time-series data (water levels, rainfall), processed tank metadata, and geospatial geometries (tank boundaries, cascade network topology).

3. Data Processing & Feature Engineering

This is where raw data is transformed into model-ready features, addressing the specific "Data Sparsity & Historical Gaps" challenge mentioned in the synopsis.

- **Spatial Interpolation & Augmentation:** Fills gaps in historical gauge and weather records using synthetic data augmentation.
- **Geospatial Processing:** Extracts tank surface area, watershed boundaries, and elevation gradients from DEM and LULC data.
- **TEK Integration Engine:** Utilizes Fuzzy Logic to convert qualitative local wisdom (e.g., "tank

4. AI/ML Analytics Core

The computational heart of the AWWE system where predictive and prescriptive insights are generated.

- **Time-Series Forecasting (LSTM / XGBoost):** Consumes historical and forecast weather/inflow data to predict future water levels and storage dynamics.
- **Spatial Classification (CNN / Random Forest):** Processes satellite imagery and elevation data to classify flood vulnerability zones and siltation levels.
- **Decision Optimization Engine:** Multi-objective algorithms that take predictive outputs and calculate the optimal trade-off between structural discharge (flood mitigation) and water retention (dry-season storage).

5. Data Serving & Presentation Layer

How the processed insights are delivered to the final users (farmers, WRD officials, local administration).

- **Backend API:** (e.g., FastAPI / Django) Serves model predictions, optimized recommendations, and aggregated spatial data to the frontend.
- **Interactive Geospatial Dashboard:** (e.g., React with Mapbox/Leaflet) A web-based UI that visualizes:
  - Real-time hydro-dynamic risk heatmaps.
  - Prioritized lists for tank and channel restoration (`vulnerability_status`).
  - Scenario simulations based on upcoming weather forecasts.

Abstract Data Flow Diagram

